

## E-commerce Product Selection Pipeline Framework v1

Goal: Build a decision support system rather than a generic information pipeline.  
Success is measured by ROI and decision quality.

Core flow:

Collect -> Extract -> Analyze -> Score -> Candidate -> Decision -> Archive -> Observe  
-> Validate -> Learn -> Update (closed loop).

Major agents:

1. Trend Agent: discover emerging products/topics.
2. Demand Agent: identify unmet user needs.
3. Competition Agent: estimate market saturation.
4. Supply Agent: evaluate sourcing feasibility and margins.
5. Risk Agent: compliance/IP/brand risks.
6. Decision Agent: aggregate evidence into candidate scores.

Definition of Done:

Produce a ranked candidate list with evidence, confidence, recommended actions, and rationale.

Validation & Learning:

Archive every prediction with evidence snapshot. Continuously collect ground-truth signals (trend, competition, supply, sales, profit). Compare predictions with reality, analyze failure causes, and update rules/models/features.